



Java Arithmetic Operators

[< Previous](#)[Next >](#)

Arithmetic Operators

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example	Try it
+	Addition	Adds together two values	$x + y$	Try it »
-	Subtraction	Subtracts one value from another	$x - y$	Try it »
*	Multiplication	Multiplies two values	$x * y$	Try it »
/	Division	Divides one value by another	x / y	Try it »
%	Modulus	Returns the division remainder	$x \% y$	Try it »
++	Increment	Increases the value of a variable by 1	<code>++x</code>	Try it »
--	Decrement	Decreases the value of a variable by 1	<code>--x</code>	Try it »

Here is an example using different arithmetic operators in one example:



```
int x = 10;
int y = 3;

System.out.println(x + y); // 13
System.out.println(x - y); // 7
System.out.println(x * y); // 30
System.out.println(x / y); // 3
System.out.println(x % y); // 1

int z = 5;
++z;
System.out.println(z); // 6
--z;
System.out.println(z); // 5
```

Try it Yourself »

Note: When dividing two integers in Java, the result will also be an integer. For example, `10 / 3` gives `3`. If you want a decimal result, use `double` values, like `10.0 / 3`.

Example

```
int a = 10;
int b = 3;
System.out.println(a / b); // Integer division, result is 3

double c = 10.0d;
double d = 3.0d;
System.out.println(c / d); // Decimal division, result is 3.333...
```



Incrementing and Decrementing

Incrementing and decrementing are very common in programming, especially when working with counters, loops, and arrays (which you will learn more about in later chapters).

The `++` operator increases a value by 1, while the `--` operator decreases a value by 1:

Example

```
int x = 5;

++x; // Increment x by 1
System.out.println(x); // 6

--x; // Decrement x by 1
System.out.println(x); // 5
```

[Try it Yourself »](#)

Real Life Example: Counting People

Imagine you are building a program to count how many people enter and leave a room. You can use `++` to increase the counter when someone enters, and `--` to decrease it when someone leaves:

Example

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

```
peopleInRoom++;  
peopleInRoom++;  
peopleInRoom++;  
  
System.out.println(peopleInRoom); // 3  
  
// 1 person leaves  
peopleInRoom--;  
  
System.out.println(peopleInRoom); // 2
```

[Try it Yourself »](#)[◀ Previous](#)[Sign in to track progress](#)[Next >](#)

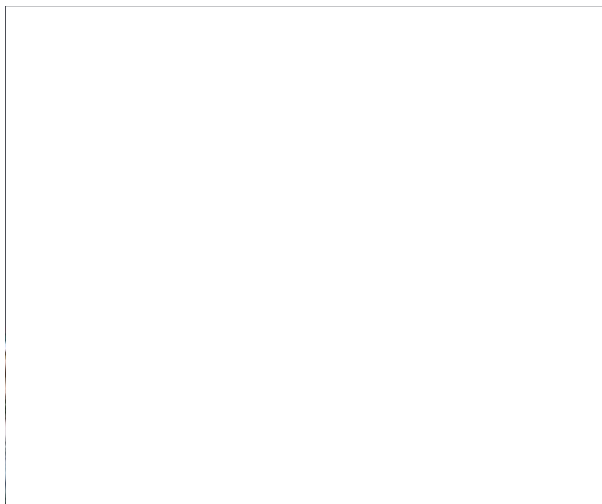
ADVERTISEMENT

[Tutorials ▼](#)[References ▼](#)[Exercises ▼](#)[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)

COLOR PICKER

[REMOVE ADS](#)

ADVERTISEMENT





GET CERTIFIED

FOR TEACHERS

FOR BUSINESS

CONTACT US

Top Tutorials

- HTML Tutorial
- CSS Tutorial
- JavaScript Tutorial
- How To Tutorial
- SQL Tutorial
- Python Tutorial
- W3.CSS Tutorial
- Bootstrap Tutorial
- PHP Tutorial
- Java Tutorial
- C++ Tutorial
- jQuery Tutorial

Top References

- HTML Reference
- CSS Reference
- JavaScript Reference
- SQL Reference
- Python Reference
- W3.CSS Reference
- Bootstrap Reference
- PHP Reference
- HTML Colors
- Java Reference
- AngularJS Reference
- jQuery Reference

Top Examples

- HTML Examples
- CSS Examples
- JavaScript Examples
- How To Examples
- SQL Examples
- Python Examples
- W3.CSS Examples
- Bootstrap Examples
- PHP Examples
- Java Examples

Get Certified

- HTML Certificate
- CSS Certificate
- JavaScript Certificate
- Front End Certificate
- SQL Certificate
- Python Certificate
- PHP Certificate
- jQuery Certificate
- Java Certificate
- C++ Certificate

[Tutorials](#) ▼[References](#) ▼[Exercises](#) ▼[Sign In](#)[CSS](#)[JAVASCRIPT](#)[SQL](#)[PYTHON](#)[JAVA](#)[PHP](#)[HOW TO](#)[W3.CSS](#)[C](#)[FORUM](#) [ABOUT](#) [ACADEMY](#)

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning.

Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our [terms of use](#), [cookie and privacy policy](#).

[Copyright 1999-2025](#) by Refsnes Data. All Rights Reserved. [W3Schools](#) is Powered by [W3.CSS](#).